

Supplement to

Plants in limbo – the theory of detrital photosynthesis

Luka Seamus Wright^{1,2*}

¹Oceans Institute, University of Western Australia, Perth, Australia

²School of Biological Sciences, University of Western Australia, Perth, Australia

*luka.wright@research.uwa.edu.au, luka@wright.it, orcid.org/0000-0002-1273-6256

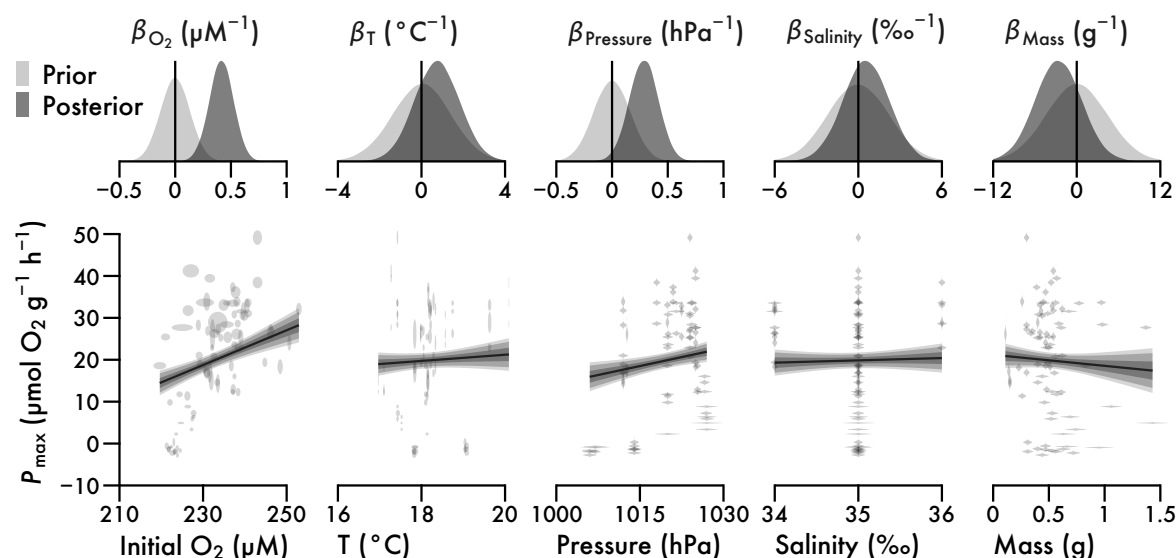


Figure 1. Effect of confounding variables associated with incubation on light-saturated net seagrass photosynthesis ($P_{\max}$) per gram of blotted mass. Distributions are kernel density estimates of priors and posteriors for slopes of the multiple linear regression. $P_{\max}$, initial oxygen (O_2) and temperature (T) are measured with error. Ellipses and violins are bi- and univariate posterior probability distributions for observations. Lines and intervals are means and the central 50, 80 and 90% of posterior probability for the mean prediction of $P_{\max}$.

Table S1. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies and observations in each non-taxonomic group are highlighted in bold.

Species		Days	Studies	Observations
Chlorophyta				
<u>Ulvales</u>				
Ulvaaceae	<i>Ulva clathrata</i>	30	1	6
	<i>Ulva intestinalis</i>	30	1	6
	<i>Ulva reticulata</i>	30	1	6
Heterokontophyta				
<u>Desmarestiales</u>				
Desmarestiaceae	<i>Desmarestia anceps</i>	313	1	65
<u>Fucales</u>				
Durvillaeaceae	<i>Durvillaea antarctica</i>	14	2	448
Sargassaceae	<i>Sargassum fallax</i>	14	1	84
	<i>Sargassum spinuligerum</i>	14	1	159
Seirococcaceae	<i>Scytothalia dorycarpa</i>	14	1	48

<u>Laminariales</u>				
Laminariaceae	<i>Laminaria digitata</i>	46	3	162
	<i>Laminaria hyperborea</i>	168	6	533
	<i>Laminaria ochroleuca</i>	56	4	416
	<i>Macrocystis pyrifera</i>	55	3	363
	<i>Saccharina latissima</i>	39	1	42
Lessoniaceae	<i>Ecklonia radiata</i>	119	2	117
<u>Tilopteridales</u>				
Phyllariaceae	<i>Saccorhiza polyschides</i>	39	1	38
Streptophyta				
<u>Alismatales</u>				
Cymodoceaceae	<i>Amphibolis antarctica</i> *	34	1	39
Hydrocharitaceae	<i>Elodea densa</i>	16	2	43
	<i>Halophila ovalis</i> *	34	1	42
	<i>Hydrilla verticillata</i>	12	7	506
	<i>Thalassia testudinum</i> *	52	1	14
	<i>Vallisneria americana</i>	104	1	5
	<i>Vallisneria spiralis</i>	9	6	59
Potamogetonaceae	<i>Potamogeton</i> sp.	104	1	5
	<i>Stuckenia pectinata</i>	9	5	55
Zosteraceae	<i>Zostera marina</i> *	85	1	10
	<i>Zostera muelleri</i> *	16	1	58
<u>Apiales</u>				
Apiaceae	<i>Coriandrum sativum</i>	9	1	24
	<i>Petroselinum crispum</i>	7	1	7
<u>Asterales</u>				
Asteraceae	<i>Chrysanthemum</i> × <i>morifolium</i>	12	1	30
	<i>Helianthus annuus</i>	0.024	1	36
	<i>Helianthus tuberosus</i>	0.026	1	36
	<i>Inula racemosa</i>	0.025	1	36
	<i>Smallanthus connatus</i>	0.026	1	36
<u>Brassicales</u>				
Brassicaceae	<i>Arabidopsis thaliana</i>	6	2	60
	<i>Brassica napus</i>	1.7	1	125
	<i>Brassica oleracea</i>	10	1	22
	<i>Nasturtium officinale</i>	5	2	39
	<i>Raphanus raphanistrum</i>	6	1	20
	<i>Sinapis alba</i>	6	1	20
Tropaeolaceae	<i>Tropaeolum majus</i>	8	2	27
<u>Caryophyllales</u>				
Amaranthaceae	<i>Amaranthus cruentus</i>	0.026	1	36
	<i>Celosia argentea</i>	0.024	1	36
	<i>Chenopodium berlandieri</i>	0.27	1	26
	<i>Gomphrena serrata</i>	0.024	1	36
	<i>Spinacia oleracea</i>	10	3	85
Polygonaceae	<i>Koenigia weyrichii</i>	0.028	1	36
<u>Cucurbitales</u>				
Cucurbitaceae	<i>Cucumis sativus</i>	13	2	13
	<i>Sicyos edulis</i>	10	1	48
<u>Ericales</u>				
Theaceae	<i>Camellia sinensis</i>	4	1	11
<u>Fabales</u>				

Fabaceae	<i>Cercis canadensis</i>	0.0052	1	576
	<i>Lathyrus oleraceus</i>	17	2	42
	<i>Phaseolus vulgaris</i>	2.1	1	10
	<i>Trifolium subterraneum</i>	7	2	25
	<i>Vigna unguiculata</i>	4	1	170
<u>Fagales</u>				
Betulaceae	<i>Betula pendula</i>	0.029	1	36
Fagaceae	<i>Quercus alba</i>	0.01	1	24
	<i>Quercus muehlenbergii</i>	0.0052	1	192
	<i>Quercus robur</i>	0.024	2	228
	<i>Quercus rubra</i>	0.01	1	24
Juglandaceae	<i>Juglans nigra</i>	0.01	1	24
<u>Lamiales</u>				
Lamiaceae	<i>Salvia officinalis</i>	11	1	8
<u>Laurales</u>				
Lauraceae	<i>Cinnamomum tamala</i>	21	1	36
<u>Pinales</u>				
Pinaceae	<i>Larix sibirica</i>	0.025	1	36
	<i>Pinus sibirica</i>	0.023	1	33
	<i>Pinus sylvestris</i>	0.022	1	33
<u>Poales</u>				
Cyperaceae	<i>Bolboschoenus fluviatilis</i>	104	1	5
Poaceae	<i>Avena sativa</i>	8	8	161
	<i>Hordeum vulgare</i>	11	13	466
	<i>Lolium multiflorum</i>	5	1	44
	<i>Lolium pratense</i>	7	5	112
	<i>Oryza sativa</i>	27	11	327
	<i>Panicum miliaceum</i>	5	2	69
	<i>Secale cereale</i>	5	2	102
	<i>Triticum aestivum</i>	10	13	646
	<i>Triticum turgidum</i>	6	1	7
	<i>Triticum vulgare</i>	6	1	48
	<i>Zea mays</i>	6	3	58
Typhaceae	<i>Typha angustifolia</i>	104	1	5
<u>Polypodiales</u>				
Nephrolepidaceae	<i>Nephrolepis exaltata</i>	10	1	24
<u>Rosales</u>				
Rosaceae	<i>Fragaria × ananassa</i>	0.0056	1	663
	<i>Malus domestica</i>	21	3	150
	<i>Malus hupehensis</i>	15	1	18
<u>Sapindales</u>				
Rutaceae	<i>Citrus × aurantium</i>	10	1	5
Sapindaceae	<i>Acer saccharinum</i>	0.25	1	36
	<i>Acer truncatum</i>	0.0052	1	192
<u>Solanales</u>				
Convolvulaceae	<i>Ipomoea batatas</i>	3	1	24
Solanaceae	<i>Nicotiana rustica</i>	10	1	180
	<i>Nicotiana tabacum</i>	9	1	24
	<i>Solanum lycopersicum</i>	5	1	192
	<i>Solanum melongena</i>	16	1	24
	<i>Solanum tuberosum</i>	0.026	1	36
<u>Vitales</u>				

References. Studies included in the meta-analysis in addition to the present study. Note that 128 references are listed because the data from three studies were not independent and thus merged.

- Albright LJ, Chocair J, Masuda K, Valdes M. 1982.** Degradation of the kelps *Macrocystis integrifolia* and *Nereocystis luetkeana* in British Columbia coastal waters. In Srivastava LM (ed) *Synthetic and Degradative Processes in Marine Macrophytes*. De Gruyter, Berlin, pp 215-234. <https://doi.org/10.1515/9783110837988-015>.
- Arguelles T, Guardiola JL. 1977.** Hormonal control of senescence in excised orange leaves. *Journal of Horticultural Science* **52**: 199–204. <https://doi.org/10.1080/00221589.1977.11514748>.
- Asim M, Guo M, Khan R et al. 2022.** Investigation of sugar signaling behaviors involved in sucrose-induced senescence initiation and progression in *N. tabacum*. *Plant Physiology and Biochemistry* **184**: 112–125. <https://doi.org/10.1016/j.plaphy.2022.05.024>.
- Barua DN. 1960.** Effect of age and carbon-dioxide concentration on assimilation by detached leaves of tea and sunflower. *The Journal of Agricultural Science* **55**: 413–421. <https://doi.org/10.1017/s0021859600023327>.
- de Bettignies F, Dauby P, Thomas F et al. 2020.** Degradation dynamics and processes associated with the accumulation of *Laminaria hyperborea* (Phaeophyceae) kelp fragments: an in situ experimental approach. *Journal of Phycology* **56**: 1481–1492. <https://doi.org/10.1111/jpy.13041>.
- Bianchi TS, Findlay S. 1991.** Decomposition of Hudson estuary macrophytes: photosynthetic pigment transformations and decay constants. *Estuaries* **14**: 65–73. <https://doi.org/10.2307/1351983>.
- Birecka H, Ireton KP, Bitonti AJ, McCann PP. 1991.** Endogenous polyamine levels and dark-induced leaf senescence. *Phytochemistry* **30**: 105–108. [https://doi.org/10.1016/0031-9422\(91\)84107-4](https://doi.org/10.1016/0031-9422(91)84107-4).
- Biswal B, Joshi PN, Kulandaivelu G. 1997.** Changes in leaf protein and pigment contents and photosynthetic activities during senescence of detached maize leaves: influence of different ultraviolet radiations. *Photosynthetica* **34**: 37–44. <https://doi.org/10.1023/a:1006855532424>.
- Biswal B, NK C, Sahu P, UC B. 1983.** Senescence of detached fern leaves. *Plant and Cell Physiology* **24**: 1203–1208. <https://doi.org/10.1093/oxfordjournals.pcp.a076634>.
- Biswal UC, Mohanty P. 1976a.** Dark stress induced senescence of detached barley leaves: II. Alteration in the absorption characteristic and photochemical activity of the chloroplasts isolated from senescing leaves. *Plant Science Letters* **7**: 371–379. [https://doi.org/10.1016/0304-4211\(76\)90094-8](https://doi.org/10.1016/0304-4211(76)90094-8).
- Biswal UC, Mohanty P. 1976b.** Aging induced changes in photosynthetic electron transport of detached barley leaves. *Plant and Cell Physiology* **17**: 323–331. <https://doi.org/10.1093/oxfordjournals.pcp.a075285>.
- Biswas AK, Choudhuri MA. 1980.** Mechanism of monocarpic senescence in rice. *Plant Physiology* **65**: 340–345. <https://doi.org/10.1104/pp.65.2.340>.
- Biswas R, Sinhababu A, Banerjee A, Kar RK. 2005.** Qualitative and quantitative effect of light on leaf senescence in a submerged aquatic weed, *Hydrilla verticillata*. *Indian Journal of Plant Physiology* **10**: 66.
- Bouma D. 1975.** Effects of some metabolic phosphorus compounds on rates of photosynthesis of detached phosphorus-deficient subterranean clover leaves. *Journal of Experimental Botany* **26**: 52–59. <https://doi.org/10.1093/jxb/26.1.52>.
- Bouma D, Dowling EJ. 1982.** The relationship between the phosphorus status of subterranean clover plants and losses in chlorophyll from detached leaves: a basis for the visual diagnosis of phosphorus deficiency. *Annals of Botany* **49**: 637–648. <https://doi.org/10.1093/oxfordjournals.aob.a086291>.

- Brady CJ, Heng FT. 1975.** Rate of protein synthesis in senescing, detached wheat leaves. *Functional Plant Biology* **2**: 163–176. <https://doi.org/10.1071/pp9750163>.
- Brouwer PEM. 1996.** Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology* **16**: 129–137. <https://doi.org/10.1007/bf02390433>.
- Causin HF, Marchetti CF, Pena LB, Gallego SM, Barneix AJ. 2015.** Down-regulation of catalase activity contributes to senescence induction in wheat leaves exposed to shading stress. *Biologia Plantarum* **59**: 154–162. <https://doi.org/10.1007/s10535-014-0480-z>.
- Chalise R, Dahal A, Basnet S, Sharma S, Pant DR, Khanal R. 2024.** Effect of plasma-activated water on chlorophyll retention in detached Tejpat (*Cinnamomum tamala*) leaves. *Heliyon* **10**: e24480. <https://doi.org/10.1016/j.heliyon.2024.e24480>.
- Chen H-J, Tsai Y-J, Chen W-S, Huang G-J, Huang S-S, Lin Y-H. 2010.** Ethephon-mediated effects on leaf senescence are affected by reduced glutathione and EGTA in sweet potato detached leaves. *Botanical Studies* **51**: 171–181.
- Claussen W, Hawker JS, Loveys BR. 1985.** Sucrose synthase activity, invertase activity, net photosynthetic rates and carbohydrate content of detached leaves of eggplants as affected by attached stems and shoots (sinks). *Journal of Plant Physiology* **119**: 123–131. [https://doi.org/10.1016/s0176-1617\(85\)80171-1](https://doi.org/10.1016/s0176-1617(85)80171-1).
- Dent RM, Cowan AK. 1994.** Absciscic acid-induced senescence in light-and dark-incubated detached leaves of barley. *South African Journal of Botany* **60**: 340–344. [https://doi.org/10.1016/s0254-6299\(16\)30589-0](https://doi.org/10.1016/s0254-6299(16)30589-0).
- Dolui TK, Jana S. 1988.** Effects of phytohormones on some biochemical parameters during dark induced leaf senescence of *Secium edule* on Darjeeling hill of the Eastern Himalayas. *Biologia Plantarum* **30**: 379–383. <https://doi.org/10.1007/bf02878194>.
- Embry JL, Nothnagel EA. 1994.** Photosynthetic light-harvesting during leaf senescence in *Panicum miliaceum*. *Plant Science* **95**: 141–152. [https://doi.org/10.1016/0168-9452\(94\)90088-4](https://doi.org/10.1016/0168-9452(94)90088-4).
- Embry JL, Nothnagel EA. 1988.** Leaf development and senescence in *Panicum miliaceum* L., a cereal with a short seed-to-seed cycle. *Plant Science* **55**: 129–136. [https://doi.org/10.1016/0168-9452\(88\)90168-9](https://doi.org/10.1016/0168-9452(88)90168-9).
- Even-Chen Z, Atsmon D, Itai C. 1978.** Hormonal aspects of senescence in detached tobacco leaves. *Physiologia Plantarum* **44**: 377–382. <https://doi.org/10.1111/j.1399-3054.1978.tb01641.x>.
- Frontier N, de Bettignies F, Foggo A, Davoult D. 2021.** Sustained productivity and respiration of degrading kelp detritus in the shallow benthos: detached or broken, but not dead. *Marine Environmental Research* **166**: 105277. <https://doi.org/10.1016/j.marenvres.2021.105277>.
- Frontier N, Mulas M, Foggo A, Smale DA. 2022.** The influence of light and temperature on detritus degradation rates for kelp species with contrasting thermal affinities. *Marine Environmental Research* **173**: 105529. <https://doi.org/10.1016/j.marenvres.2021.105529>.
- Gapper NE, Coupe SA, McKenzie MJ et al. 2005.** Senescence-associated down-regulation of 1-aminocyclopropane-1-carboxylate (ACC) oxidase delays harvest-induced senescence in broccoli. *Functional Plant Biology* **32**: 891–901. <https://doi.org/10.1071/fp05076>.
- Gauthier M-M, Jacobs DF. 2018.** Reductions in net photosynthesis and stomatal conductance vary with time since leaf detachment in three deciduous angiosperms. *Trees* **32**: 1247–1252. <https://doi.org/10.1007/s00468-018-1706-z>.
- Gi E, Cho S-H, Kim S-H, Kang K, Paek N-C. 2024.** Rice ONAC016 promotes leaf senescence through abscisic acid signaling pathway involving OsNAP. *The Crop Journal* <https://doi.org/10.1016/j.cj.2024.02.009>.
- Grover A, Sabat SC, Mohanty P. 1986.** Effect of temperature on photosynthetic activities of senescing detached wheat leaves. *Plant and Cell Physiology* **27**: 117–126. <https://doi.org/10.1093/oxfordjournals.pcp.a077071>.

- Grover A, Sabat SC, Mohanty P. 1987.** Does the loss of leaf chlorophyll during senescence of primary wheat leaf arise due to loss in chloroplast number or chlorophyll content. *Biochemie und Physiologie der Pflanzen* **182**: 481–484. [https://doi.org/10.1016/s0015-3796\(87\)80078-1](https://doi.org/10.1016/s0015-3796(87)80078-1).
- Gut H, Rutz C, Matile P, Thomas H. 1987.** Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: degradation of carotenoids. *Physiologia Plantarum* **70**: 659–663. <https://doi.org/10.1111/j.1399-3054.1987.tb04321.x>.
- Hilditch P, Thomas H, Rogers L. 1986.** Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: photosynthesis and photosynthetic electron transport. *Planta* **167**: 146–151. <https://doi.org/10.1007/bf00446382>.
- Housley TL, Pollock CJ. 1985.** Photosynthesis and carbohydrate metabolism in detached leaves of *Lolium temulentum* L. *New Phytologist* **99**: 499–507. <https://doi.org/10.1111/j.1469-8137.1985.tb03678.x>.
- Huang J, Han B, Xu S, Zhou M, Shen W. 2011.** Heme oxygenase-1 is involved in the cytokinin-induced alleviation of senescence in detached wheat leaves during dark incubation. *Journal of Plant Physiology* **168**: 768–775. <https://doi.org/10.1016/j.jplph.2010.10.010>.
- Huang M, Feng L, He Y, Zhu Z. 2006.** Study on changing rules of chlorophyll concentration of detached canola leaves. In Xu K, Luo Q, Xing D, Priezzhev AV, Tuchin VV (eds) Fourth International Conference on Photonics and Imaging in Biology and Medicine, 6047. SPIE, Tianjin, pp 604740. <https://doi.org/10.1117/12.710949>.
- Hukmani P, Tripathy BC. 1994.** Chlorophyll biosynthetic reactions during senescence of excised barley (*Hordeum vulgare* L. cv IB 65) leaves. *Plant Physiology* **105**: 1295–1300. <https://doi.org/10.1104/pp.105.4.1295>.
- Jana S, Choudhuri MA. 1979.** Photosynthetic, photorespiratory and respiratory behaviour of three submersed aquatic angiosperms. *Aquatic Botany* **7**: 13–19. [https://doi.org/10.1016/0304-3770\(79\)90003-2](https://doi.org/10.1016/0304-3770(79)90003-2).
- Jana S, Choudhuri MA. 1980.** Senescence in submerged aquatic angiosperms: changes in intact and isolated leaves during aging. *New Phytologist* **86**: 191–198. <https://doi.org/10.1111/j.1469-8137.1980.tb03188.x>.
- Jana S, Choudhuri MA. 1982a.** Changes occurring during aging and senescence in a submerged aquatic angiosperm (*Potamogeton pectinatus*). *Physiologia Plantarum* **55**: 356–360. <https://doi.org/10.1111/j.1399-3054.1982.tb00305.x>.
- Jana S, Choudhuri MA. 1982b.** Glycolate metabolism of three submersed aquatic angiosperms during ageing. *Aquatic Botany* **12**: 345–354. [https://doi.org/10.1016/0304-3770\(82\)90026-2](https://doi.org/10.1016/0304-3770(82)90026-2).
- Jana S, Choudhuri MA. 1982c.** Changes in the activities of ribulose 1, 5-bisphosphate and phosphoenolpyruvate carboxylases in submersed aquatic angiosperms during aging. *Plant Physiology* **70**: 1125–1127. <https://doi.org/10.1104/pp.70.4.1125>.
- Jana S, Choudhuri MA. 1984a.** Synergistic effects of heavy metal pollutants on senescence in submerged aquatic plants. *Water, Air, & Soil Pollution* **21**: 351–357. <https://doi.org/10.1007/bf00163635>.
- Jana S, Choudhuri MA. 1984b.** Effects of plant growth regulators on Hill activity of submerged aquatic plants during induced senescence. *Aquatic Botany* **18**: 371–376. [https://doi.org/10.1016/0304-3770\(84\)90057-3](https://doi.org/10.1016/0304-3770(84)90057-3).
- Jana S. 1988.** Glycolate metabolism of wheat and rice leaves during senescence and under the influence of growth regulators. *Biologia Plantarum* **30**: 30–33. <https://doi.org/10.1007/bf02876420>.
- Jia HS, Han YQ, Li DQ. 2003.** Photoinhibition and active oxygen species production in detached apple leaves during dehydration. *Photosynthetica* **41**: 151–156. <https://doi.org/10.1023/a:1025889219115>.

- Jiang W, Sheng Q, Zhou X-J, Zhang M-J, Liu X-J. 2002.** Regulation of detached coriander leaf senescence by 1-methylcyclopropene and ethylene. *Postharvest Biology and Technology* **26**: 339–345. [https://doi.org/10.1016/s0925-5214\(02\)00068-6](https://doi.org/10.1016/s0925-5214(02)00068-6).
- Kang K, Kim YS, Park S, Back K. 2009.** Evaluation of light-harvesting complex proteins as senescence-related protein markers in detached rice leaves. *Photosynthetica* **47**: 638–640. <https://doi.org/10.1007/s11099-009-0093-5>.
- Kao CH, Yang SF. 1983.** Role of ethylene in the senescence of detached rice leaves. *Plant Physiology* **73**: 881–885. <https://doi.org/10.1104/pp.73.4.881>.
- Kar M, Feierabend J. 1984.** Metabolism of activated oxygen in detached wheat and rye leaves and its relevance to the initiation of senescence. *Planta* **160**: 385–391. <https://doi.org/10.1007/bf00429753>.
- Kar M, Streb P, Hertwig B, Feierabend J. 1993.** Sensitivity to photodamage increases during senescence in excised leaves. *Journal of Plant Physiology* **141**: 538–544. [https://doi.org/10.1016/s0176-1617\(11\)80453-0](https://doi.org/10.1016/s0176-1617(11)80453-0).
- Kar RK, Choudhuri MA. 1985.** Senescence of *Hydrilla* leaves in light and darkness. *Physiologia Plantarum* **63**: 225–230. <https://doi.org/10.1111/j.1399-3054.1985.tb01907.x>.
- Kar S, Montague DT, Villanueva-Morales A. 2021a.** Measurement of photosynthesis in excised leaves of ornamental trees: a novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees* **35**: 889–905. <https://doi.org/10.1007/s00468-021-02088-w>.
- Kar S, Montague T, Villanueva-Morales A, Hellman E. 2021b.** Measurement of gas exchange on excised grapevine leaves does not differ from *in situ* leaves, and potentially shortens sampling time. *Applied Sciences* **11**: 3644. <https://doi.org/10.3390/app11083644>.
- Karatas I, Ozturk L, Ersahin Y, Okatan Y. 2010.** Effects of auxin on photosynthetic pigments and some enzyme activities during dark-induced senescence of *Tropaeolum* leaves. *Pakistan Journal of Botany* **42**: 1881–1888.
- Kato MC, Hikosaka K, Hirose T. 2002.** Leaf discs floated on water are different from intact leaves in photosynthesis and photoinhibition. *Photosynthesis Research* **72**: 65–70. <https://doi.org/10.1023/a:1016097312036>.
- Knauer GA, Ayers AV. 1977.** Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography* **22**: 408–414. <https://doi.org/10.4319/lo.1977.22.3.0408>.
- Kreslavski VD, Lankin AV, Vasilyeva GK et al. 2014.** Effects of polyaromatic hydrocarbons on photosystem II activity in pea leaves. *Plant Physiology and Biochemistry* **81**: 135–142. <https://doi.org/10.1016/j.plaphy.2014.02.020>.
- Landolt R, Matile P. 1990.** Glyoxisome-like microbodies in senescent spinach leaves. *Plant Science* **72**: 159–163. [https://doi.org/10.1016/0168-9452\(90\)90078-3](https://doi.org/10.1016/0168-9452(90)90078-3).
- Lewington RJ, Talbot M, Simon EW. 1967.** The yellowing of attached and detached cucumber cotyledons. *Journal of Experimental Botany* **18**: 526–534. <https://doi.org/10.1093/jxb/18.3.526>.
- Lewington RJ, Simon EW. 1969.** The effect of light on the senescence of detached cucumber cotyledons. *Journal of Experimental Botany* **20**: 138–144. <https://doi.org/10.1093/jxb/20.1.138>.
- Liu L, Li H, Zeng H, Cai Q, Zhou X, Yin C. 2016.** Exogenous jasmonic acid and cytokinin antagonistically regulate rice flag leaf senescence by mediating chlorophyll degradation, membrane deterioration, and senescence-associated genes expression. *Journal of Plant Growth Regulation* **35**: 366–376. <https://doi.org/10.1007/s00344-015-9539-0>.
- Lovell PH, Illsley ANN, Moore KG. 1972.** The effects of light intensity and sucrose on root formation, photosynthetic ability, and senescence in detached cotyledons of *Sinapis alba* L.

- and *Raphanus sativus* L. *Annals of Botany* **36**: 123–134.
<https://doi.org/10.1093/oxfordjournals.aob.a084565>.
- Martin C, Thimann KV. 1972.** The role of protein synthesis in the senescence of leaves: I. The formation of protease. *Plant Physiology* **49**: 64–71. <https://doi.org/10.1104/pp.49.1.64>.
- Miller BL, Huffaker RC. 1985.** Differential induction of endoproteinases during senescence of attached and detached barley leaves. *Plant Physiology* **78**: 442–446.
<https://doi.org/10.1104/pp.78.3.442>.
- Mishra D, Kar M. 1973.** Effect of benzimidazole and nickel ions in the control of *Oryza sativa* leaf senescence by red and far-red light. *Phytochemistry* **12**: 1521–1522.
[https://doi.org/10.1016/0031-9422\(73\)80360-7](https://doi.org/10.1016/0031-9422(73)80360-7).
- Misr AN, Meena M. 1986.** Effect of temperature on senescing detached rice leaves: I. Photoelectron transport activity of chloroplasts. *Plant Science* **46**: 1–4.
[https://doi.org/10.1016/0168-9452\(86\)90123-8](https://doi.org/10.1016/0168-9452(86)90123-8).
- Misra AN, Biswal UC. 1980.** Effect of phytohormones on chlorophyll degradation during aging of chloroplasts in vivo and in vitro. *Protoplasma* **105**: 1–8.
<https://doi.org/10.1007/bf01279845>.
- Misra AN, Biswal UC. 1982.** Differential changes in the electron transport properties of thylakoid membranes during aging of attached and detached leaves, and of isolated chloroplasts. *Plant, Cell & Environment* **5**: 27–30. <https://doi.org/10.1111/1365-3040.ep11587573>.
- Momose T, Ozeki Y. 2013.** Regulatory effect of stems on sucrose-induced chlorophyll degradation and anthocyanin synthesis in *Egeria densa* leaves. *Journal of Plant Research* **126**: 859–867. <https://doi.org/10.1007/s10265-013-0581-3>.
- Mondal R, Choudhuri MA. 1981.** Role of hydrogen peroxide in senescence of excised leaves of rice and maize. *Biochemie und Physiologie der Pflanzen* **176**: 700–709.
[https://doi.org/10.1016/s0015-3796\(81\)80055-8](https://doi.org/10.1016/s0015-3796(81)80055-8).
- Montague T, Villanueva-Morales A, Khan MAR, Wallace R, Panta S. 2024.** Comparing the accuracy and efficiency of leaf gas exchange measurements of excised and nonexcised strawberry (*Fragaria × ananassa* Duch. ‘Camino Real’) leaves. *HortTechnology* **34**: 728–737. <https://doi.org/10.21273/horttech05516-24>.
- Nedunchezian N, Ravindran KC, Kulandaivelu G. 1995.** Changes in photosynthetic apparatus during dark incubation of detached leaves from control and ultraviolet-B treated *Vigna* seedlings. *Biologia Plantarum* **37**: 341–348. <https://doi.org/10.1007/bf02913238>.
- Nock LP, Rogers LJ, Thomas H. 1992.** Metabolism of protein and chlorophyll in leaf tissue of *Festuca pratensis* during chloroplast assembly and senescence. *Phytochemistry* **31**: 1465–1470. [https://doi.org/10.1016/0031-9422\(92\)83088-g](https://doi.org/10.1016/0031-9422(92)83088-g).
- Ogwen J-O, Song X-S, Hu W-H, Shi K, Zhou Y-H, Yu J-Q. 2009.** Detached leaves of tomato differ in their photosynthetic physiological response to moderate high and low temperature stress. *Scientia Horticulturae* **123**: 17–22.
<https://doi.org/10.1016/j.scienta.2009.07.011>.
- Oh M-H, Moon Y-H, Lee C-H. 2003.** Increased stability of LHCII by aggregate formation during dark-induced leaf senescence in the *Arabidopsis* mutant, ore10. *Plant and Cell Physiology* **44**: 1368–1377. <https://doi.org/10.1093/pcp/pcg170>.
- Okada K, Inoue Y, Satoh K, Katoh S. 1992.** Effects of light on degradation of chlorophyll and proteins during senescence of detached rice leaves. *Plant and Cell Physiology* **33**: 1183–1191. <https://doi.org/10.1093/oxfordjournals.pcp.a078372>.
- Pellikaan GC. 1982.** Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological Bulletin* **16**: 83–92. <https://doi.org/10.1007/bf02255416>.
- Philosoph-Hadas S, Meir S, Akiri B, Kanner J. 1994a.** Oxidative defense systems in leaves of three edible herb species in relation to their senescence rates. *Journal of Agricultural and Food Chemistry* **42**: 2376–2381. <https://doi.org/10.1021/jf00047a003>.

- Philosoph-Hadas S, Meir S, Aharoni N. 1991.** Effect of wounding on ethylene biosynthesis and senescence of detached spinach leaves. *Physiologia Plantarum* **83**: 241–246. <https://doi.org/10.1111/j.1399-3054.1991.tb02148.x>.
- Philosoph-Hadas S, Meir S, Aharoni N. 1994b.** Role of ethylene in senescence of watercress leaves. *Physiologia Plantarum* **90**: 553–559. <https://doi.org/10.1111/j.1399-3054.1994.tb08814.x>.
- Reyes-Arribas T, Barrett JE, Huber DJ, Nell TA, Clark DG. 2001.** Leaf senescence in a non-yellowing cultivar of chrysanthemum (*Dendranthema grandiflora*). *Physiologia Plantarum* **111**: 540–544. <https://doi.org/10.1034/j.1399-3054.2001.1110415.x>.
- Richardson SD. 1957.** The effect of leaf age on the rate of photosynthesis in detached leaves of tree seedlings. *Acta Botanica Neerlandica* **6**: 445–457. <https://doi.org/10.1111/j.1438-8677.1957.tb00592.x>.
- Rodoni S, Schellenberg M, Matile P. 1998.** Chlorophyll breakdown in senescing barley leaves as correlated with phaeophorbide oxygenase activity. *Journal of Plant Physiology* **152**: 139–144. [https://doi.org/10.1016/s0176-1617\(98\)80124-7](https://doi.org/10.1016/s0176-1617(98)80124-7).
- Rodríguez MT, González MP, Linares JM. 1987.** Degradation of chlorophyll and chlorophyllase activity in senescing barley leaves. *Journal of Plant Physiology* **129**: 369–374. [https://doi.org/10.1016/s0176-1617\(87\)80094-9](https://doi.org/10.1016/s0176-1617(87)80094-9).
- Rothäusler E, Gómez I, Hinojosa IA et al. 2011.** Kelp rafts in the Humboldt Current: Interplay of abiotic and biotic factors limit their floating persistence and dispersal potential. *Limnology and Oceanography* **56**: 1751–1763. <https://doi.org/10.4319/lo.2011.56.5.1751>.
- Sabater B, Rodríguez MT. 1978.** Control of chlorophyll degradation in detached leaves of barley and oat through effect of kinetin on chlorophyllase levels. *Physiologia Plantarum* **43**: 274–276. <https://doi.org/10.1111/j.1399-3054.1978.tb02577.x>.
- Sağlam-Çağ S. 2007.** The effect of epibrassinolide on senescence in wheat leaves. *Biotechnology & Biotechnological Equipment* **21**: 63–65. <https://doi.org/10.1080/13102818.2007.10817415>.
- Shafique S, Siddiqui PJS, Aziz RA, Burhan Z, Mansoor SN. 2010.** Weight loss and changes in organic, inorganic and chlorophyll contents in three species of seaweeds during decomposition. *Pakistan Journal of Botany* **42**: 2599–2604.
- Sharma R, Biswal UC. 1976.** Effect of kinetin and light on peroxidase activity of detached barley leaves. *Zeitschrift für Pflanzenphysiologie* **78**: 169–172. [https://doi.org/10.1016/s0044-328x\(78\)80188-3](https://doi.org/10.1016/s0044-328x(78)80188-3).
- Shaw M, Bhattacharya PK, Quick WA. 1965.** Chlorophyll, protein, and nucleic acid levels in detached, senescing wheat leaves. *Canadian Journal of Botany* **43**: 739–746. <https://doi.org/10.1139/b65-083>.
- Singh N, Mishra D. 1975.** The effect of benzimidazole and red light on the senescence of detached leaves of *Oryza sativa* cv. Ratna. *Physiologia Plantarum* **34**: 67–74. <https://doi.org/10.1111/j.1399-3054.1975.tb01858.x>.
- Špundová M, Strzałka K, Nauš J. 2005.** Xanthophyll cycle activity in detached barley leaves senescing under dark and light. *Photosynthetica* **43**: 117–124. <https://doi.org/10.1007/s11099-005-7124-7>.
- Špundová M, Popelková H, Ilík P, Skotnica J, Novotný R, Nauš J. 2003.** Ultra-structural and functional changes in the chloroplasts of detached barley leaves senescing under dark and light conditions. *Journal of Plant Physiology* **160**: 1051–1058. <https://doi.org/10.1078/0176-1617-00902>.
- Srivastava BI, Ware G. 1965.** The effect of kinetin on nucleic acids and nucleases of excised barley leaves. *Plant Physiology* **40**: 62. <https://doi.org/10.1104/pp.40.1.62>.
- Strother S, Vatta R. 1986.** The effects of light, cycloheximide and incubation medium on the senescence of detached seagrass leaves. *Biologia Plantarum* **28**: 211–218. <https://doi.org/10.1007/bf02894599>.

- Subhan D, Murthy SDS. 2000.** Synergistic effect of AlCl₃ and kinetin on chlorophyll and protein contents and photochemical activities in detached wheat primary leaves during dark incubation. *Photosynthetica* **38**: 211–214. <https://doi.org/10.1023/a:1007261614200>.
- Subhan D, Murthy SDS. 2001a.** Senescence retarding effect of metal ions: pigment and protein contents and photochemical activities of detached primary leaves of wheat. *Photosynthetica* **39**: 53–58. <https://doi.org/10.1023/a:1012487718114>.
- Subhan D, Murthy SDS. 2001b.** Effect of polyamines on chlorophyll and protein contents, photochemical activity, and energy transfer in detached wheat leaves during dark incubation. *Biologia Plantarum* **44**: 529–533. <https://doi.org/10.1023/a:1013734502112>.
- Tala F, Velásquez M, Mansilla A, Macaya EC, Thiel M. 2016.** Latitudinal and seasonal effects on short-term acclimation of floating kelp species from the South-East Pacific. *Journal of Experimental Marine Biology and Ecology* **483**: 31–41. <https://doi.org/10.1016/j.jembe.2016.06.003>.
- Tala F, Penna-Díaz MA, Luna-Jorquera G, Rothäusler E, Thiel M. 2017.** Daily and seasonal changes of photobiological responses in floating bull kelp *Durvillaea antarctica* (Chamisso) Hariot (Fucales: Phaeophyceae). *Phycologia* **56**: 271–283. <https://doi.org/10.2216/16-93.1>.
- Tetley RM, Thimann KV. 1974.** The metabolism of oat leaves during senescence: I. Respiration, carbohydrate metabolism, and the action of cytokinins. *Plant Physiology* **54**: 294–303. <https://doi.org/10.1104/pp.54.3.294>.
- Thimann KV, Tetley RR, Van Thanh T. 1974.** The metabolism of oat leaves during senescence: II. Senescence in leaves attached to the plant. *Plant Physiology* **54**: 859–862. <https://doi.org/10.1104/pp.54.6.859>.
- Thimann KV, Tetley RM, Krivak BM. 1977.** Metabolism of oat leaves during senescence: V. Senescence in light. *Plant Physiology* **59**: 448–454. <https://doi.org/10.1104/pp.59.3.448>.
- Thimann KV, Satler SO. 1979.** Relation between leaf senescence and stomatal closure: senescence in light. *Proceedings of the National Academy of Sciences* **76**: 2295–2298. <https://doi.org/10.1073/pnas.76.5.2295>.
- Thimann KV. 1985.** The senescence of detached leaves of *Tropaeolum*. *Plant Physiology* **79**: 1107–1110. <https://doi.org/10.1104/pp.79.4.1107>.
- Thomas H. 1987.** Sid: a Mendelian locus controlling thylakoid membrane disassembly in senescing leaves of *Festuca pratensis*. *Theoretical and Applied Genetics* **73**: 551–555. <https://doi.org/10.1007/bf00289193>.
- Thomas H, Stoddart JL. 1975.** Separation of chlorophyll degradation from other senescence processes in leaves of a mutant genotype of meadow fescue (*Festuca pratensis* L.). *Plant Physiology* **56**: 438–441. <https://doi.org/10.1104/pp.56.3.438>.
- Turner WB, Bidwell RGS. 1965.** Rates of photosynthesis in attached and detached bean leaves, and the effect of spraying with indoleacetic acid solution. *Plant Physiology* **40**: 446. <https://doi.org/10.1104/pp.40.3.446>.
- van Hees DH, Olsen YS, Mattio L, Ruiz-Montoya L, Wernberg T, Kendrick GA. 2019.** Cast adrift: physiology and dispersal of benthic *Sargassum spinuligerum* in surface rafts. *Limnology and Oceanography* **64**: 526–540. [10.1002/lno.11057](https://doi.org/10.1002/lno.11057).
- Veierskov B, Kirk HG. 1986.** Senescence in oat leaf segments under hypobaric conditions. *Physiologia Plantarum* **66**: 283–287. <https://doi.org/10.1111/j.1399-3054.1986.tb02421.x>.
- Veierskov B. 1987.** Irradiance-dependent senescence of isolated leaves. *Physiologia Plantarum* **71**: 316–320. <https://doi.org/10.1111/j.1399-3054.1987.tb04349.x>.
- Vlčková A, Špundová M, Kotabová E, Novotný R, Doležal K, Nauš J. 2006.** Protective cytokinin action switches to damaging during senescence of detached wheat leaves in continuous light. *Physiologia Plantarum* **126**: 257–267. <https://doi.org/10.1111/j.1399-3054.2006.00593.x>.

- Voronin PY, Fedoseeva GP. 2012.** Stomatal control of photosynthesis in detached leaves of woody and herbaceous plants. *Russian Journal of Plant Physiology* **59**: 281–286. <https://doi.org/10.1134/s1021443712020173>.
- Wang P, Yin L, Liang D, Li C, Ma F, Yue Z. 2012.** Delayed senescence of apple leaves by exogenous melatonin treatment: toward regulating the ascorbate-glutathione cycle. *Journal of Pineal Research* **53**: 11–20. <https://doi.org/10.1111/j.1600-079x.2011.00966.x>.
- Wen B, Li C, Fu X et al. 2019.** Effects of nitrate deficiency on nitrate assimilation and chlorophyll synthesis of detached apple leaves. *Plant Physiology and Biochemistry* **142**: 363–371. <https://doi.org/10.1016/j.plaphy.2019.07.007>.
- Wright LS, Foggo A. 2021.** Photosynthetic pigments of co-occurring Northeast Atlantic *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series* **678**: 227–232. <https://doi.org/10.3354/meps13886>.
- Wright LS, Pessarrodona A, Foggo A. 2022.** Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. *Global Change Biology* **28**: 5514–5531. <https://doi.org/10.1111/gcb.16299>.
- Wright LS, Kregting L. 2023.** Genus-specific response of kelp photosynthetic pigments to decomposition. *Marine Biology* **170**: 144. <https://doi.org/10.1007/s00227-023-04289-y>.
- Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T. 2024.** Temperature sensitivity of detrital photosynthesis. *Annals of Botany* **133**: 17–28. <https://doi.org/10.1093/aob/mcad167>.
- Xiao X, de Bettignies T, Olsen YS, Agusti S, Duarte CM, Wernberg T. 2015.** Sensitivity and acclimation of three canopy-forming seaweeds to UVB radiation and warming. *PLoS One* **10**: e0143031. <https://doi.org/10.1371/journal.pone.0143031>.
- Yang J, Worley E, Udvardi M. 2014.** A NAP-AAO3 regulatory module promotes chlorophyll degradation via ABA biosynthesis in *Arabidopsis* leaves. *The Plant Cell* **26**: 4862–4874. <https://doi.org/10.1105/tpc.114.133769>.
- Yoshida Y. 1970.** Control of chloroplast senescence of detached *Elodea* leaves. *Shokubutsugaku Zasshi* **83**: 137–143. <https://doi.org/10.15281/jplantres1887.83.137>.
- Zhang Z, Li G, Gao H et al. 2012.** Characterization of photosynthetic performance during senescence in stay-green and quick-leaf-senescence *Zea mays* L. inbred lines. *PLoS One* **7**: e42936. <https://doi.org/10.1371/journal.pone.0042936>.
- Zhang Z, Guo J, Yang C et al. 2023.** Exogenous dopamine delays the senescence of detached *Malus hupehensis* leaves by reducing phytohormone signalling and sugar degradation. *Scientia Horticulturae* **319**: 112151. <https://doi.org/10.1016/j.scienta.2023.112151>.
- Zhao L, Zhang H, Zhang B, Bai X, Zhou C. 2012.** Physiological and molecular changes of detached wheat leaves in responding to various treatments. *Journal of Integrative Plant Biology* **54**: 567–576. <https://doi.org/10.1111/j.1744-7909.2012.01139.x>.